Nishal Stanislaus Silva, *PhD*

Research Scientist | Real-Time Multimodal ML, Benchmarks & Open Datasets

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Research scientist in real-time multimodal machine learning, with a PhD (University of Trento, 2025) and 10+ years across industry and academia. My work fuses and models heterogeneous sensor streams under hard timing constraints: inertial, audio and mixed-reality tracking data synchronized and classified in real time, with hierarchical state machines governing transitions between behavior modes. These are the sensor-fusion, latency, behavior-modeling and safe-deployment problems of autonomy perception, applied in a different domain. My published models run inference in 14 ms on embedded hardware, F1 0.76, 74× faster than the baseline (JAES 2026); a gesture-and-audio fusion model reaches F1 0.78 across a 500-event corpus (IEEE IS² 2025). I have designed four open benchmark datasets, published 10+ peer-reviewed papers, and reviewed for IEEE venues.

Core Competencies

ML Research: multimodal learning and sensor fusion; sequence and time-series modelling (RNN, LSTM, GRU), CNNs, hierarchical state machines, imitation-style rule models; generative modelling (VAE, diffusion) for low-label regimes; benchmark, ablation and dataset design; statistical evaluation; linear algebra, calculus, probability
Systems: Python, C++17; real-time inference under sub-30 ms budgets; multi-stream synchronization and alignment; edge and server deployment with monitoring; profiling; reproducible, tested pipelines
Research practice: 10+ peer-reviewed publications; four open datasets (70+ contributors); reviewer and programme committee for IEEE IS² and IEEE I3DA; student supervision; EU-consortium project delivery

Technical Stack

Languages: Python, C++17, C, MATLAB, JavaScript/TypeScript, Shell
ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, ONNX; RNN/CNN, VAE and diffusion prototypes, structural-similarity and synthetic-variation methods; NumPy, Pandas, SciPy
Systems & Data: real-time streaming pipelines, multi-sensor time synchronization, Docker, Git, pytest, CI/CD, Airflow, AWS; Raspberry Pi / Embedded Linux, IMU / I²C, Unity + Meta XR SDK

Education

Ph.D, Information Engineering & Computer Science – *University of Trento* *Jan 2025*
Thesis: Embedded Real-time Musical Pattern Detection for Smart Musical Instruments

M.Sc, Telecommunication & Electronic Engineering – *Sheffield Hallam University* *Aug 2018*

B.Eng (Hons), Electronic Engineering – *Sheffield Hallam University* *Mar 2014*

Research & Work Experience

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 – Dec 2025*

- Built a real-time multimodal pipeline synchronizing audio, inertial, neural (EEG) and mixed-reality head- and hand-tracking streams across four participants simultaneously, with SAE Institute Barcelona and g.tec; EU Horizon EIC Pathfinder project MUSMET.

- Designed real-time sequence models (stacked RNNs, single-LSTM classifiers) and the baseline and ablation suites quantifying architecture, feature and latency trade-offs; 14 ms inference on embedded hardware, F1 0.76, 74× faster than DTW (JAES 2026).
- Four peer-reviewed papers in 2025; supervised two BSc thesis students; coordinated across a 12-partner research consortium.

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 – Aug 2024*

- Fused inertial (IMU) and audio streams through a hierarchical finite state machine (Idle to Pattern to Gesture), reaching P 0.79 / R 0.76 / F1 0.78 across a 500-event corpus (IEEE IS² 2025).
- Profiled compute and power for real-time feasibility on self-powered embedded compute; explored VAE- and diffusion-based synthetic data for limited-label regimes.

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 – Mar 2024*

- Ran a human-centred evaluation of an ML-driven interactive system, combining quantitative metrics with structured interviews to produce usability and adoption findings (IEEE I3DA 2025).

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 – Jul 2020*

- Built end-to-end computer-vision and sensor systems for manufacturing, integrating camera and IoT streams into production workflows with real-time C++/Python inference on embedded hardware.
- Designed minimal-data detection and registration algorithms; automated inspection cut QC time 99.5% and lifted throughput up to 300%; introduced CI/CD and code review; mentored engineers.

Publications (selected; full list: *nishal.xyz/publications*)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. J. Audio Eng. Soc. 74(3), Mar. 2026.
- Silva, N. and Turchet, L. *Smart Drums: Comparing Audio and MIDI in Embedded Real-Time Drum Pattern Recognition*. J. Audio Eng. Soc., 2026 (in press).
- Silva, N. and Turchet, L. *Music Information Retrieval as a Service: Latency Characterization of an Edge-to-Server Architecture*. IEEE Int. Symp. on the Internet of Sounds, 2026 (accepted).
- Silva, N., Wanderley, M. and Turchet, L. *Melody and Motion: Integrating Guitar Gestures with Musical Patterns for Extended Control*. IEEE Int. Symp. on the Internet of Sounds, Oct. 2025.
- Silva, N., Boem, A. and Turchet, L. *Interactive IoMusT-Based Concerts: Real-Time Pattern Recognition and Audience Experience*. IEEE Int. Conf. on Immersive and 3D Audio, Sep. 2025.
- Silva, N. and Turchet, L. *Real-Time Pattern Recognition of Symbolic Monophonic Music*. 19th Int. Audio Mostly Conference (ACM), Sep. 2024.
- Silva, N., Weeraddana, P.C. and Fischione, C. *On Musical Onset Detection via the S-Transform*. 52nd Asilomar Conference on Signals, Systems, and Computers, Oct. 2018.

Datasets, Service & Recognition

- **Open datasets:** DoMP, DoPP, DoDP, DoDP2 – 9,800+ time-aligned multi-take recordings from 70+ contributors, published on Zenodo as real-time detection benchmarks.
- **Peer review:** Reviewer for IEEE Access and IEEE I3DA; programme committee for IEEE International Symposium on the Internet of Sounds.
- **MIDI Innovation Awards:** Finalist (Sep 2023), Prototypes & Non-Commercial Software category, for a real-time detection system on embedded hardware.